

PRIME PALINDROME

Java Programming Assignment

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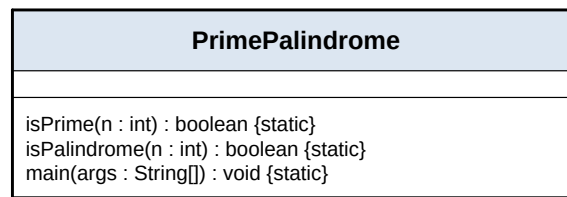
1. ALGORITHM

Steps for executing the Prime Palindrome program:

Step	Algorithm
1	Start the program.
2	Import the Scanner class.
3	Define isPrime(n) to check whether the given number is prime.
4	If n is less than 2, return false.
5	Check divisibility from 2 to n/2. If n is divisible by any value, return false; otherwise return true.
6	Define isPalindrome(n) and store the original number.
7	Reverse the number by repeatedly extracting the last digit and building the reversed number.
8	Compare the original number with the reversed number and return true if they are equal.
9	Create a Scanner object and read the number from the user.
10	Check both isPrime(n) and isPalindrome(n).
11	If both conditions are true, display that the number is a Prime Palindrome; otherwise display that it is not.
12	Close the Scanner and stop the program.

2. UML CLASS DIAGRAM

The PrimePalindrome class contains two static checking methods and the main() method.



contains

uses both methods

UML: The isPrime() and isPalindrome() methods are static and are called from main(). Scanner is used for input.

3. JAVA SOURCE CODE

File name: **PrimePalindrome.java**

```
import java.util.Scanner;

public class PrimePalindrome {

    // Method to check prime
    static boolean isPrime(int n) {
        if (n < 2)
            return false;

        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0)
                return false;
        }
        return true;
    }

    // Method to check palindrome
    static boolean isPalindrome(int n) {
        int original = n;
        int reverse = 0;

        while (n > 0) {
            int digit = n % 10;
            reverse = reverse * 10 + digit;
            n = n / 10;
        }

        return original == reverse;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

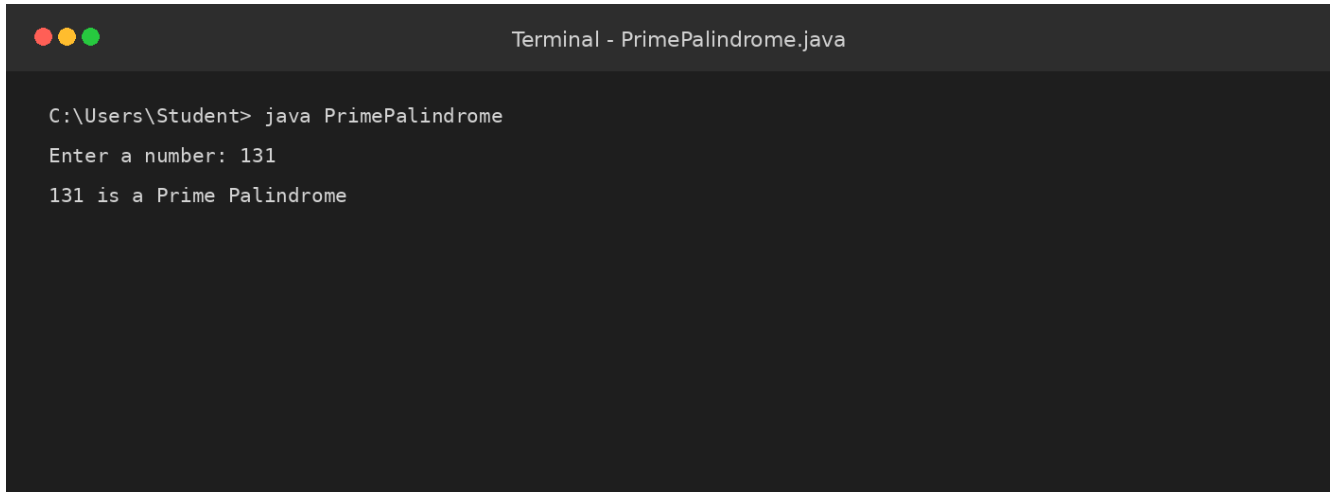
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        if (isPrime(n) && isPalindrome(n))
            System.out.println(n + " is a Prime Palindrome");
        else
            System.out.println(n + " is not a Prime Palindrome");

        sc.close();
    }
}
```

4. OUTPUT SCREEN - VS CODE STYLE

Example execution for the input **131**:

A screenshot of a terminal window titled "Terminal - PrimePalindrome.java". The terminal shows the command "C:\Users\Student> java PrimePalindrome" being executed. The program prompts "Enter a number: 131" and then outputs "131 is a Prime Palindrome".

```
Terminal - PrimePalindrome.java

C:\Users\Student> java PrimePalindrome
Enter a number: 131
131 is a Prime Palindrome
```

Result: 131 is a Prime Palindrome.